Townsend 2.1-2.4

Questions for Gallicchio

how is $|\psi\rangle\langle\psi|$ different from $\langle\psi||\psi\rangle$? If its $1\times N$ vs $N\times 1$, we still just end up with one number when doing matrix multiplication? How does the $|+z\rangle\langle+z|+|-z\rangle\langle-z|$ multiply out to make the identity?

2.1

We define the adjoint which takes conjugate transpose (hermitian?) †. This take $|\rangle \to \langle|$

We introduce the rotation generator operator $\hat{j}$ which takes infitesimal rotations

$$\hat{R}(d\phi\mathbf{k}) = 1 - \frac{i}{\hbar}\hat{J}_z d\phi$$

Which is self adjoint, and satisfies

$$\hat{R}^\dagger(d\phi\mathbf{k})\hat{R}(d\phi\mathbf{k}) = \left(1+\frac{i}{\hbar}\hat{J}_z^\dagger d\phi\right)\left(1-\frac{i}{\hbar}\hat{J}_z d\phi\right)$$

The operator undoes itself when acting with argument angles rotated by π.
Taking $N\to\infty$ actions of this $\hat{J}_z$ rotation generator, we can deflect by any angle?

Doing the rotation to make an overall phase doesn't change the state, i.e.

$$\hat{J}_z|z\rangle = |z\rangle$$

This introduces Eigenkets.

We have projection operators $\hat{P}_\pm$ that compose to make an identity operator.

$$|\psi\rangle = (\underbrace{|+z\rangle\langle+z|}_{\hat{P}_+} + \underbrace{|-z\rangle\langle-z|}_{\hat{P}_-})|\psi\rangle$$

Where each projection finds the component along a vector

$$\hat{P}_+|\psi\rangle = |+z\rangle\,\langle+z|\psi\rangle$$

This is the component of psi that lies along the z direction for the +z ket.

Blocking a path in the Modified Stern Gerlach that recombines beams is the same as taking the projection, and just leaving one portion.

We can write these as matrix representations, so

$$\vec{P}_+ = \begin{bmatrix}1 & 0\\ 0 & 0\end{bmatrix}, \vec{P}_- = \begin{bmatrix}0 & 0\\ 0 & 1\end{bmatrix}$$

or the rotation generator

$$\hat{J}_z = \begin{bmatrix}\frac{\hbar}{2} & 0\\ 0 & \frac{\hbar}{2}\end{bmatrix}$$

$$\text{if } \hat{A}|\psi\rangle = \phi$$
$$\left\langle\chi|\hat{A}|\psi\right\rangle = \langle\chi|\phi\rangle$$
$$\left\langle\psi|\hat{A}^\dagger|\chi\right\rangle = \langle\phi|\chi\rangle$$